

A Model Learned Here, Used There

The conceptual problem behind the solar work: what **SBPO 2026** settled about the local data budget, what it could not settle, and **where a formal ontology enters, why it must be a well-founded one, and what it is supposed to buy.**

Rodrigo Huber Marques Moreira Mendes · Fernanda Baião · Reinaldo Castro Souza

PUC-Rio, Industrial Engineering | 20 Brazilian MERRA-2 sites, 2017–2020 | 10 slides

Brazil measures the sun in a few places

A handful of well-instrumented sites have years of clean records. Vast regions have almost nothing.

The pragmatic move

Transfer learning. Train where data is plentiful, reuse where it is not. For the national grid operator this is the difference between forecasting a new plant and waiting years to instrument it.

The two questions it hides

- (1) May I reuse this model here at all?
- (2) How much local data before adapting it beats deploying it as-is?

WHY THIS IS NOT JUST ENGINEERING Reuse presupposes that the target is, **in the respects that govern predictive performance, the same kind of place** as the source. That is not a statistical primitive — it is a claim about which properties of a site are essential to its solar behaviour and which are incidental.

Roadmap: **I.** the problem → **II.** what SBPO 2026 did, and why that design → **III.** what it settled → **IV.** what it could not settle → **V.** where ontology enters → **VI.** for what — and the test it already failed.

Measure the budget, end to end, at 19 targets

“How Much Local Data Is Enough? Few-Shot Transfer Learning for Solar Irradiation Forecasting Across Brazil”

Piece	As built
Data	MERRA-2 reanalysis via Renewables Ninja, 2017–2020, hourly → daily means, 20 locations , 2,805,120 points. Four features: air temperature, cloud coverage, TOA irradiation, GHI (daylight only). Last 365 days held out .
Geography	Source GIG / Galeão . Ten intra-urban Rio sites, plus a north-northwest transect at ~220 km intervals out to Miranda do Norte, Maranhão .
Model	Random Forest, hyperparameters from Bayesian optimisation with 3-fold TimeSeriesSplit CV.
Strategies	ZTL zero-shot source · FTL residual corrector on top of it · TO target-only, no source knowledge.
Brackets	ZTLB = accuracy at zero local data cost · CEIL = full-train, best obtainable with unlimited local data.
Sweep	28 window sizes per site — 1–15 days and 1–13 months, always the most-recent k, temporal order preserved. MAPE on the shared hold-out; SHAP TreeExplainer for attribution.

THE TWO TARGETS **k***, the crossover window where the adapted model overtakes the unadapted source; and **k****, the ceiling-reach window where it matches a model trained on three full years of local data.

Why the design is built around what an operator can see

The best yardstick in the experiment is the one no deployment ever has

The constraint that shapes everything

CEIL is unobservable in operation. It requires years of local data — exactly the scarcity that motivated transfer learning in the first place. So the paper asks a deliberately narrower question: which **observable** signals — ZTLB, and the two learning curves $\text{MAPE}_{\text{TO}}(k)$ and $\text{MAPE}_{\text{FTL}}(k)$ — reproduce the CEIL-dependent behaviour?

Headroom, the yardstick

$\Delta_{\text{head}} = \text{MAPE}_{\text{ZTLB}} - \text{MAPE}_{\text{CEIL}}$ — the maximum benefit adaptation could possibly deliver. Positive at **16 of 19** sites, from **+0.0012** (Maracanã) to **+0.0442** (Cariparé).

Three inverted sites

Barra da Tijuca **−0.0012**, Viçosa **−0.0038**, Governador Valadares **−0.0045**: the source model already *beats* the locally trained ceiling there.

Hold this. Headroom is the paper's own “operationally sufficient statistic” for the benefit regime — and it is defined by a quantity you cannot have before you instrument the site.

Two benefit regimes, and one universal landmark **SETTLED**

Results that hold across 19 climatically diverse sites and need no knowledge of the ceiling

Stability regime — the Rio sites ($\Delta \leq 0.004$)

Target-only collapses: MAPE_{TO} exceeds ZTLB by **1.49–3.54×** across all 19 sites (1.92–2.78× within Rio). The residual corrector **prevents that collapse** but the absolute gain over the source stays below 0.004. Peak RTB of **55–58%** reflects how bad TO is, not how much was won.

Adaptation regime — distant sites ($\Delta \geq 0.024$)

Here residual FTL genuinely closes the gap: absolute MAPE reductions of **0.024–0.044**, crossover **within days**. Begins around **661 km** (Jequitaí, $\Delta = +0.030$), where the source's climatological prior is misaligned enough for correction to be worth something.

Coherent from day one, and a five-month ceiling on the wait

At **k = 1 day** FTL is already coherent at every site but two, while MAPE_{TO} exceeds 0.22 at 18 of 19. Santa Maria da Vitória (1,051 km): one day takes MAPE from ZTLB **0.1072** to **0.0920**, a **14.2%** gain. And **every positive-headroom site crossed over by five months** — 15 of 16, Maranhão needing seven — with FTL/TO converging to 1.0 between **6 and 10 months**: activate, monitor, retire.

Distance is a proxy, not a predictor OPEN

The paper says so itself — and that is where the conceptual problem re-opens

Site	Distance from GIG	What it does
Viçosa	230 km	Headroom inverted (−0.0038) — near, and the source already beats its local ceiling
Governador Valadares	457 km	Headroom inverted (−0.0045)
Jequitaiá	661 km	Third-highest absolute RTB in the whole experiment
Cariparé	far transect	Largest headroom of all, +0.0442

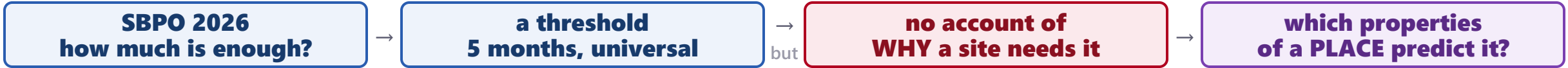
The circularity

The statistic that actually predicts the regime is **headroom** — and headroom needs **CEIL**, which needs the years of local data the whole exercise exists to avoid. The paper's honest answer is a *proxy*: $ZTLB < 0.12$ together with persistent TO instability beyond 30 days flags the zero-shot-sufficient regime. A proxy tells you **that** a site behaves a certain way. It does not tell you **what about the site** makes it so.

AND THE PAPER ASKS FOR IT Its fourth stated limitation: no formal explainability layer — “**neuro-symbolic semantic layers**” are named explicitly — has been applied, and that is called **a necessary step before automated model activation in grid-critical applications**.

The question stops being “how much” and becomes “why this much, *here*”

Everyone invokes source–target similarity. Nobody says which similarity.



What the literature supplies

Transfer works when the domains are “similar”, and diverging along the dimensions that matter produces **negative transfer**. True, and cited in the paper — but **“similar” is left as an undischarged term**, doing predictive work it was never defined precisely enough to do.

What an operator actually has on day zero

No local GHI history. What they *do* have is **categorical knowledge about the place**: its climate class, its biome, its latitude and altitude, its distance. Inherited scientific classifications, available before a single sensor is installed.

So the question is ontological, not statistical. Two stations may share a latitude but differ in cloud regime; share a biome but differ in altitude; sit far apart yet behave identically. **Which of those similarities licenses reuse, and which are mirages?**

A symbolic layer between the model and the place THE PROPOSAL

Not decoration on a machine-learning pipeline — a component with a job, sitting where the decision is made



Why *well-founded*, not an ad-hoc feature set

A gate is only as good as the categories it reasons with. Köppen classes, biome schemes and bioclimatic distances **already are domain ontologies** — structured vocabularies for organizing the world. A well-founded grounding is what makes their commitments **explicit, semantically precise, inspectable and independently auditable** instead of implicit in a feature vector.

Three commitments it has to get right

- ▶ **Roles, not kinds.** Source and target are relational roles a site plays in a transfer pair, not intrinsic types of station.
- ▶ **Graded relations** mediate them — similarity is not a boolean.
- ▶ **A verdict is a defeasible epistemic judgment**, the best conclusion licensed by categorical evidence — not a fact about the world. New local observations can, and should, defeat it.

What the ontology is supposed to buy

Each candidate notion of sameness becomes a distinct, falsifiable hypothesis about what licenses transfer

1 · Governance

A model that can state where it should **not** be trusted. This is SBPO 2026's own fourth limitation, and the precondition it names for **automated activation in grid-critical applications**.

2 · Testability

Every claim of the form “X-similarity licenses transfer” stated precisely enough to be **refuted by the measured budgets**. Ontology generates the hypotheses; the transfer experiments adjudicate.

3 · An audit trail

A per-criterion record of *why* a pair was judged as it was — explanatory value that survives **independently of whether the prediction holds**.

Candidate criterion	Notion of “sameness”	Ontological source
Radiative compatibility	Comparable top-of-atmosphere forcing	Astronomy / site geometry
Seasonal alignment	Matching shape of the annual cycle	Climatology
Climate identity	Same Köppen class	Köppen–Geiger classification
Ecological proximity	Near in biome / climate space	Whittaker biome scheme
Geographic nearness	Small great-circle distance	Spatial proximity

WHY KEEP THEM SEPARATE These are **distinct relations that often disagree** — SBPO 2026 already showed the last one failing on its own. Bundled into one “similarity score” they cannot be told apart.

And the commitment already paid its own way — by failing

The first operationalization was put under severe test, and the measured budgets refuted it

The gate's verdict does not order the data budget **FALSIFIED**

A leave-one-source-out re-analysis over all $20 \times 19 = 380$ directed pairs, under the strict permanence rule: Mann–Whitney $U = 27.0$, $p = 1.0$ — the statistic lands *exactly* on its null expectation, and a 1035-combination threshold sweep found nothing in the hypothesised direction. What replaced it: the **zero-shot baseline error** schedules the budget, confirmed across five independent machineries, bounded by **75% residual variance**.

This is the argument, not a setback

An ad-hoc feature set could not have failed this way, because it never said anything refutable. **The commitment was stated precisely enough to be killed by evidence — and it was.**

The thesis

Transferability is a structured, investigable phenomenon, and formal ontology has a concrete, testable role in explaining it — a source of hypotheses a learning system can evaluate, not scaffolding assumed in advance.

Beyond solar. A source model, a new context, an adaptation budget, rival similarity relations — the same skeleton runs through hydrology, air quality and agriculture. The contribution is a **way of asking**.